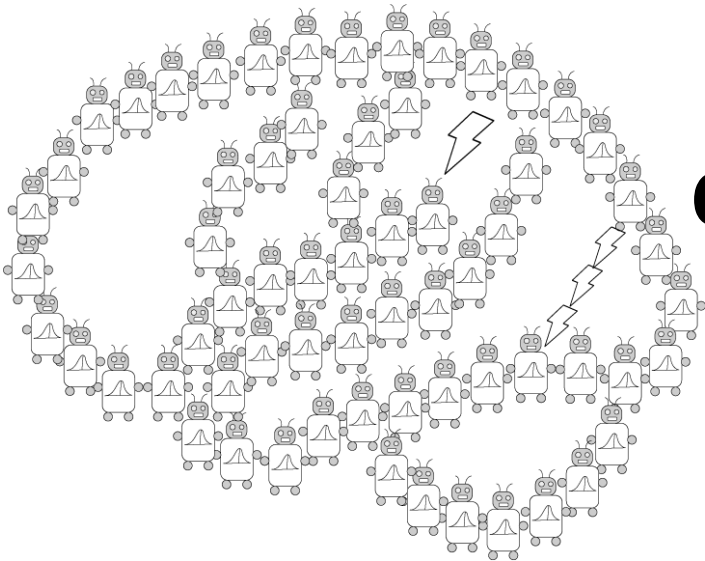


Approximation: Monte Carlo



**PSYCH/COMP SCI/BMI 841:
Computational Cognitive Science**

Joseph Austerweil

March 13, 2025

- Admin Matters
- Why Approximate?
- Monte Carlo
 - Basic sampling
 - Importance sampling
 - Sequential importance sampling
 - (aka particle filtering)
- **Short rest 1**
- Markov Chain Monte Carlo
- **Short rest 2**
- Two student presentations:
 1. Vul et al 2009 by Martin
 2. Sanborn & Griffiths 2007 by Anirudh

Administrative Matters

- Generalization Assignment grading a bit delayed.
 - Metareviews for cogsci blindsided me.
- Thank you for turning in the project proposal.
 - Will go through these next week
- Deadlines for assignments 3 and 4 pushed back a week:
 - Assignment 3 is due April 9 at 8:59pm
 - Assignment 4 is due April 23 at 8:59pm
- Questions?

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Marr's three levels

Computation

↑
↓
constrains

“What is the goal of the computation, why is it appropriate, and what is the logic of the strategy by which it can be carried out?”

Representation and algorithm

↑
↓
constrains

“What is the representation for the input and output, and the algorithm for the transformation?”

Implementation

“How can the representation and algorithm be realized physically?”

A top-down approach...

The goal is approximating Bayesian inference.

1. Adopt good methods for solving Bayesian inference from computer science and statistics.
2. See if these methods can be interpreted as psychological processes.

The result: “rational process models”

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The Monte Carlo principle

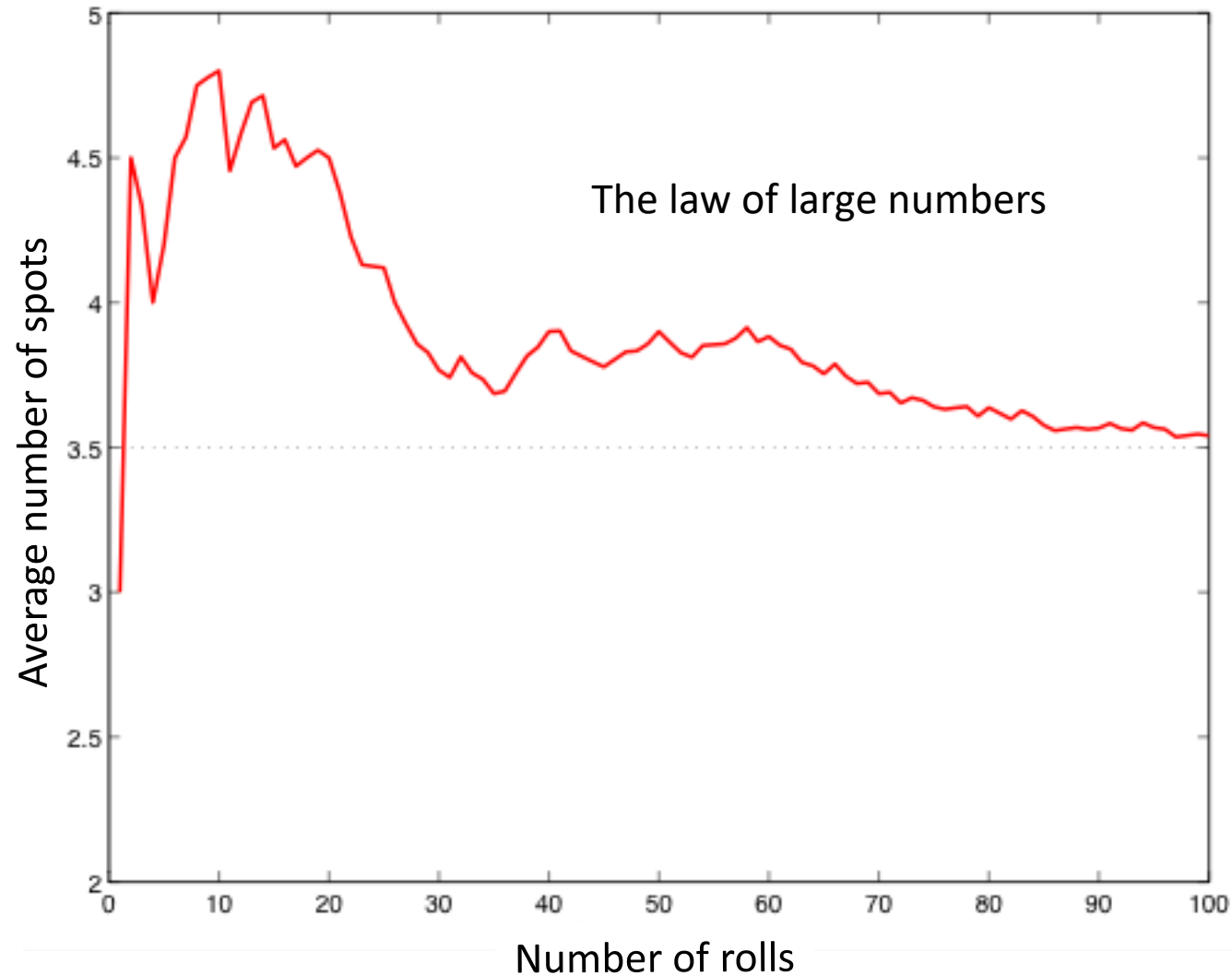
The expectation of f with respect to P can be approximated by

$$E_{P(x)}[f(x)] \approx \frac{1}{n} \sum_{i=1}^n f(x_i)$$

where the x_i are sampled from $P(x)$.

Example: the average # of spots on a die roll

The Monte Carlo principle



More formally
(and words you might read)

$$\mu = E_{P(x)}[f(x)] \approx \frac{1}{n} \sum_{i=1}^n f(x_i) = \mu_{MC}$$

μ_{MC} is consistent, $(\mu_{MC} - \mu) \rightarrow 0$ a.s. as $n \rightarrow \infty$

μ_{MC} is unbiased, with $E[\mu_{MC}] = \mu$

μ_{MC} is asymptotically normal*, with

$$\sqrt{n}(\mu_{MC} - \mu) \rightarrow N(0, \sigma_{MC}^2) \quad \text{in distribution}$$

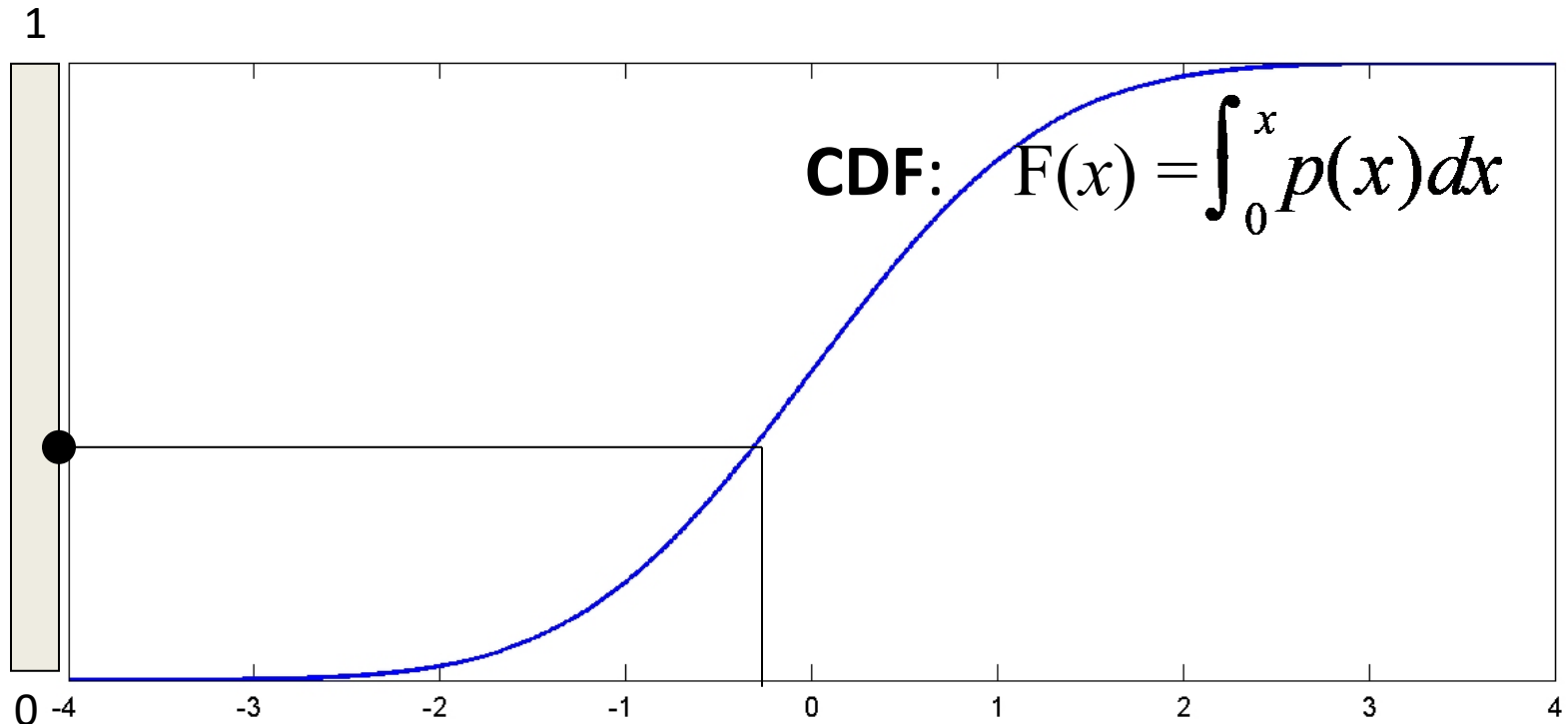
$$\sigma_{MC}^2 = E_{P(x)}[(f(x) - E_{P(x)}[f(x)])^2]$$

BUT, many times we can't sample
from the distribution.

What do we do then?

Inverse cumulative distribution

1. Draw a sample y from $\text{Uniform}(0,1)$
2. Find the value x of the Cumulative Distribution Function (CDF) such that $F(x) = y$. (i.e., $x = F^{-1}(y)$)
3. x is a sample from $p(x)$.



(requires CDF be invertible)

Example sampling via inverse cumulative distribution function (CDF)

How to sample **Exponential** random variables?

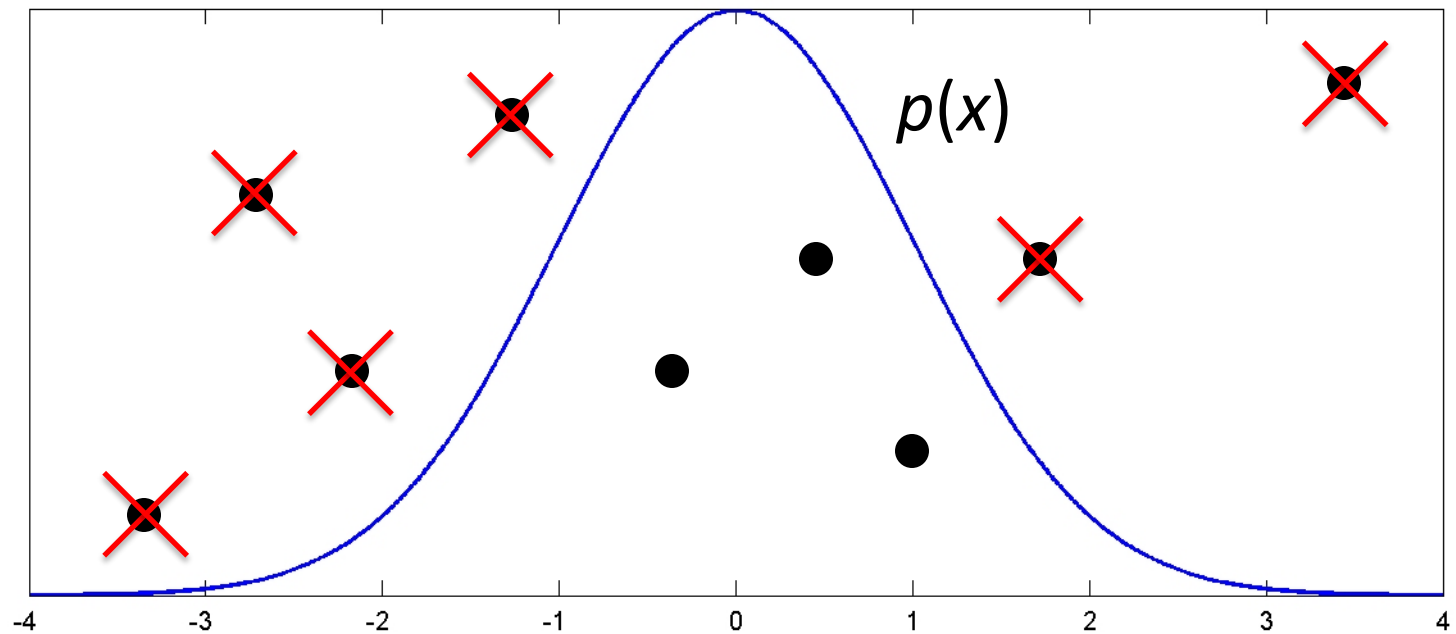
$$X \sim \text{Exp}(1) \Rightarrow p(x) = e^{-x}$$

Inverse CDF Method works well for this!

Is this better than importance sampling?

Why yes or why not?

Rejection sampling



Calculating Pi with Rejection Sampling

Area is πr^2 . If r is 1, then area of circle is π .

Sample $X, Y \sim \text{Uniform}[0,1]$

T times

What proportion of X & Y fall within circle?

$$\frac{1}{T} \sum I \left(\sqrt{x^2 + y^2} \leq 1 \right)$$

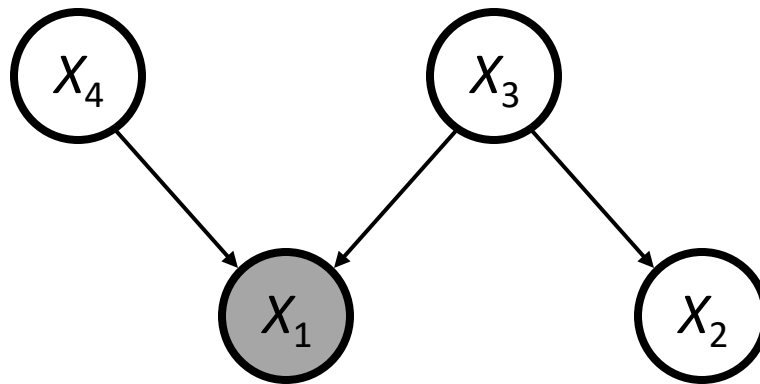
$1/4^{\text{th}}$ area of circle, π estimate is 4 times this proportion

Rejection sampling of the posterior

Generate samples of all variables following the generative process in the model.

This follows the joint distribution.

Reject all samples that do not match the observed data.



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Importance sampling

A general scheme for sampling from complex distributions that have simpler relatives.

Simple methods for sampling from posterior distributions in some cases (easy to sample from prior, prior and posterior are close).

Can be more efficient than simple Monte Carlo.

Links to exemplar models in psychology.

Also provides a solution to the question of how people can update beliefs as data come in...

Importance sampling

Basic idea:

1. generate from the wrong distribution q .*
2. assign weights to samples to correct for this.

$$\begin{aligned} E_{p(x)}[f(x)] &= \int f(x)p(x)dx \\ &= \int f(x)\frac{p(x)}{q(x)}q(x)dx \\ &\approx \frac{1}{n} \sum_{i=1}^n f(x_i) \frac{p(x_i)}{q(x_i)} \quad \text{for } x_i \sim q(x) \end{aligned}$$

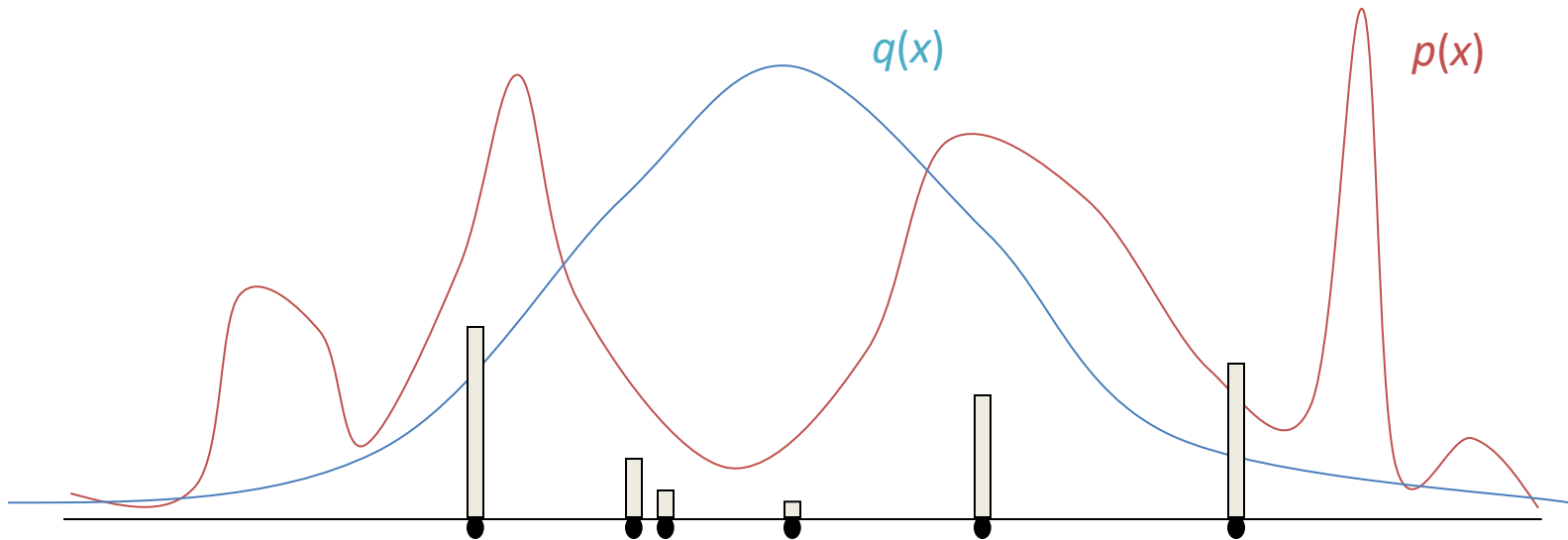
Importance sampling

$$E_{p(x)}[f(x)] \approx \frac{1}{n} \sum_{i=1}^n f(x_i) \frac{p(x_i)}{q(x_i)} \quad \text{for } x_i \sim q(x)$$

$$E_{p(x)}[f(x)] \approx \frac{\sum_{i=1}^n f(x_i) \frac{p(x_i)}{q(x_i)}}{\sum_{i=1}^n \frac{p(x_i)}{q(x_i)}} \quad \text{for } x_i \sim q(x)$$

when $p(x)$ is only known up to a multiplicative constant.

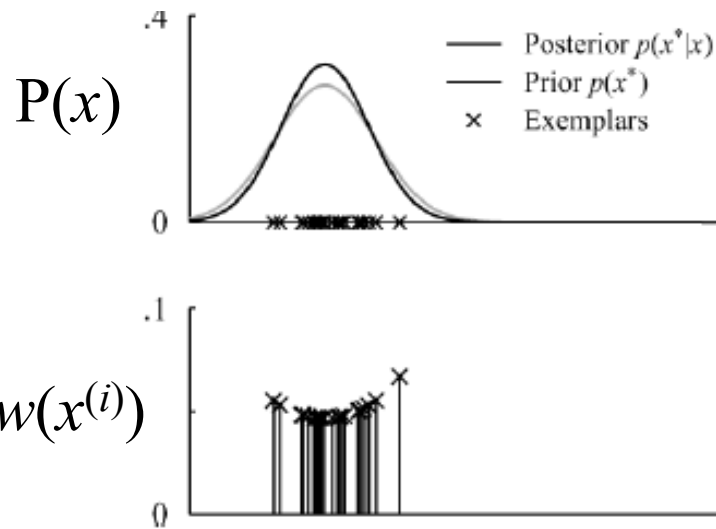
Importance sampling



$$w^{(i)} = \frac{p(x^{(i)})}{q(x^{(i)})} \quad E_p[f(x)] \approx \frac{\sum_i w^{(i)} f(x^{(i)})}{\sum_i w^{(i)}}$$

Importance sampling from prior

Overlap



Importance sampling (properties)

μ_{IS} is consistent, $(\mu_{IS} - \mu) \rightarrow 0$ a.s. as $n \rightarrow \infty$

μ_{IS} is asymptotically normal, with

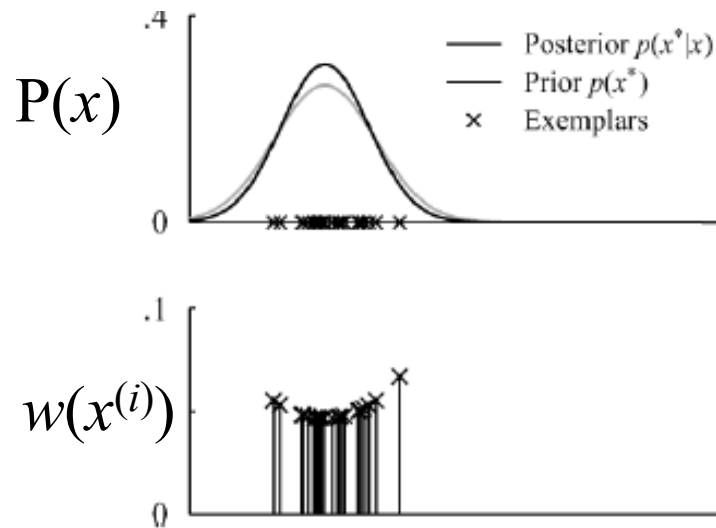
$$\sigma_{IS}^2 = E_{p(x)} \left[(f(x) - E_{p(x)}[f(x)])^2 \frac{p(x)}{q(x)} \right]$$

μ_{IS} is biased, with

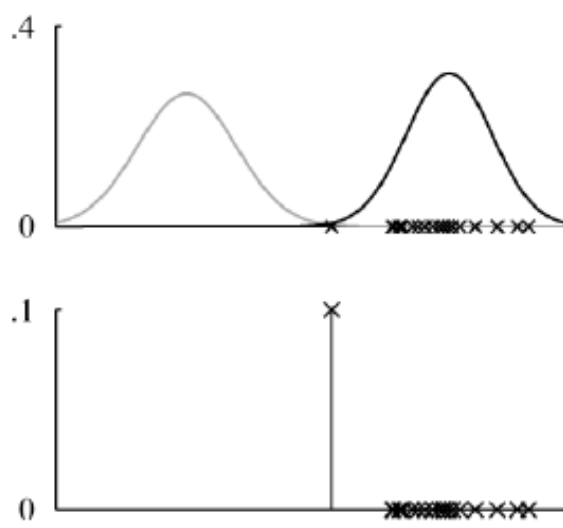
$$\mu_{IS} - \mu = \frac{1}{n} \left(E_{p(x)}[f(x)] E_{p(x)} \left[\frac{p(x)}{q(x)} \right] - E_{p(x)} \left[f(x) \frac{p(x)}{q(x)} \right] \right)$$

Importance sampling from prior

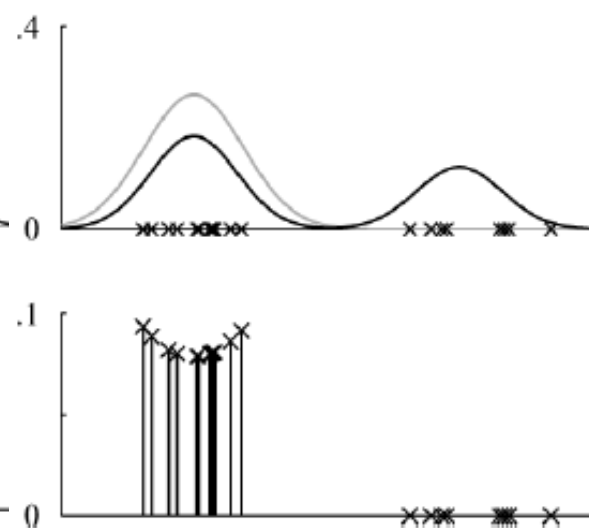
Overlap



No overlap



Some overlap



Variance of weights gives idea of how *good* the sampler is...

Optimal importance sampling

Asymptotic variance is

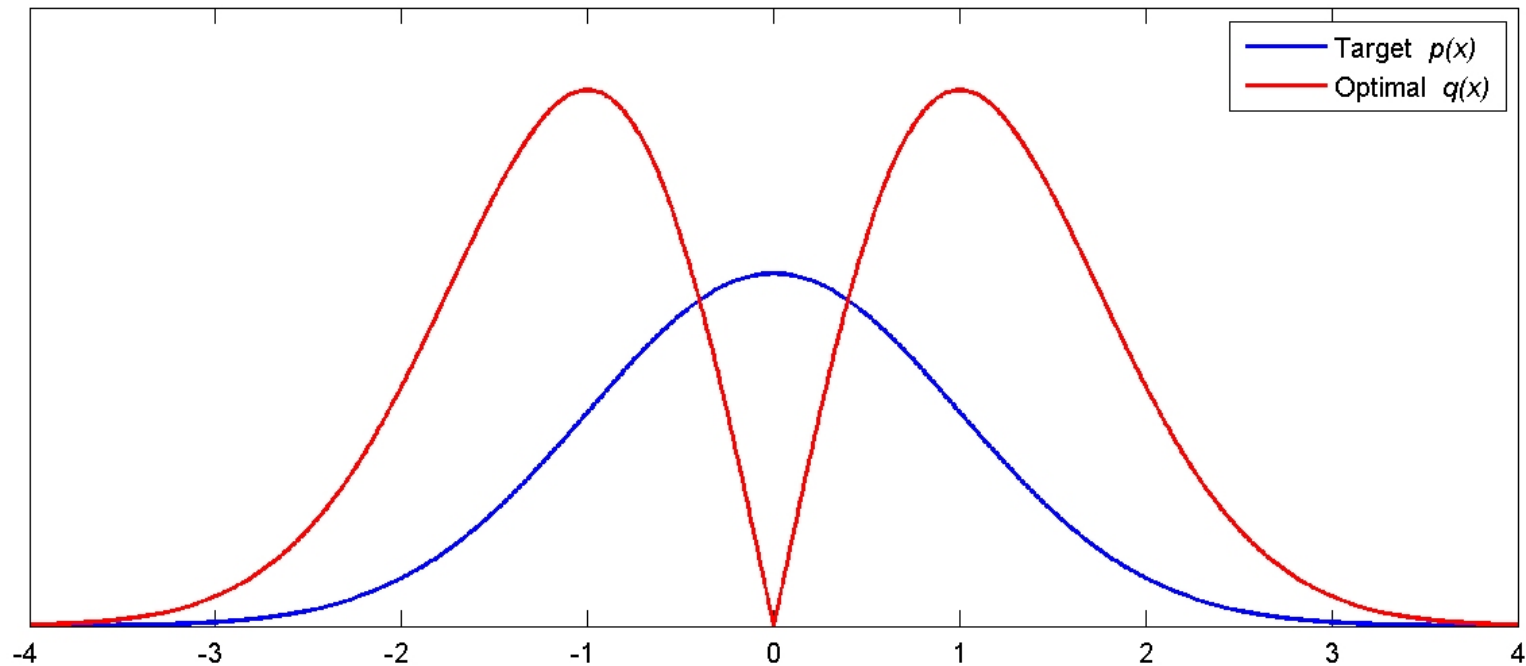
$$\sigma_{IS}^2 = E_{p(x)} \left[(f(x) - E_{p(x)}[f(x)])^2 \frac{p(x)}{q(x)} \right]$$

This is minimized by

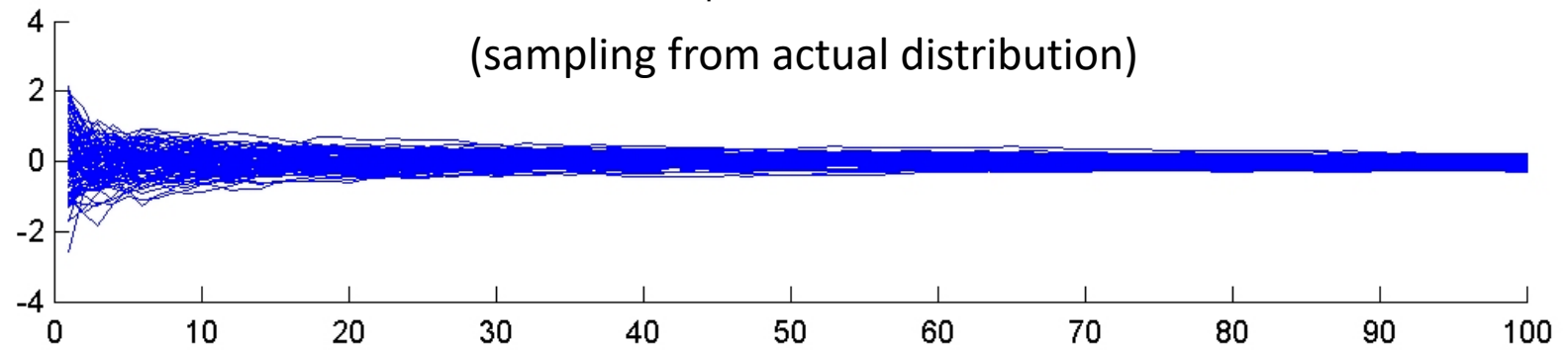
$$q(x) \propto |f(x) - E_{p(x)}[f(x)]| p(x)$$

Optimal importance sampling

Task: Estimate the mean and variance of a normal distribution. What's the optimal distribution to sample from?



Simple Monte Carlo
(sampling from actual distribution)



Likelihood weighting

Assume we only know what likelihood is prop. to.

A particularly simple form of importance sampling for **posterior distributions** ($p(h|x)$).

Use the prior as the proposal distribution ($q=p(h)$).

Weights:

$$w = \frac{p(h|d)}{p(h)} = \frac{p(d|h)p(h)}{p(d)p(h)} = \frac{p(d|h)}{p(d)} \propto p(d|h)$$

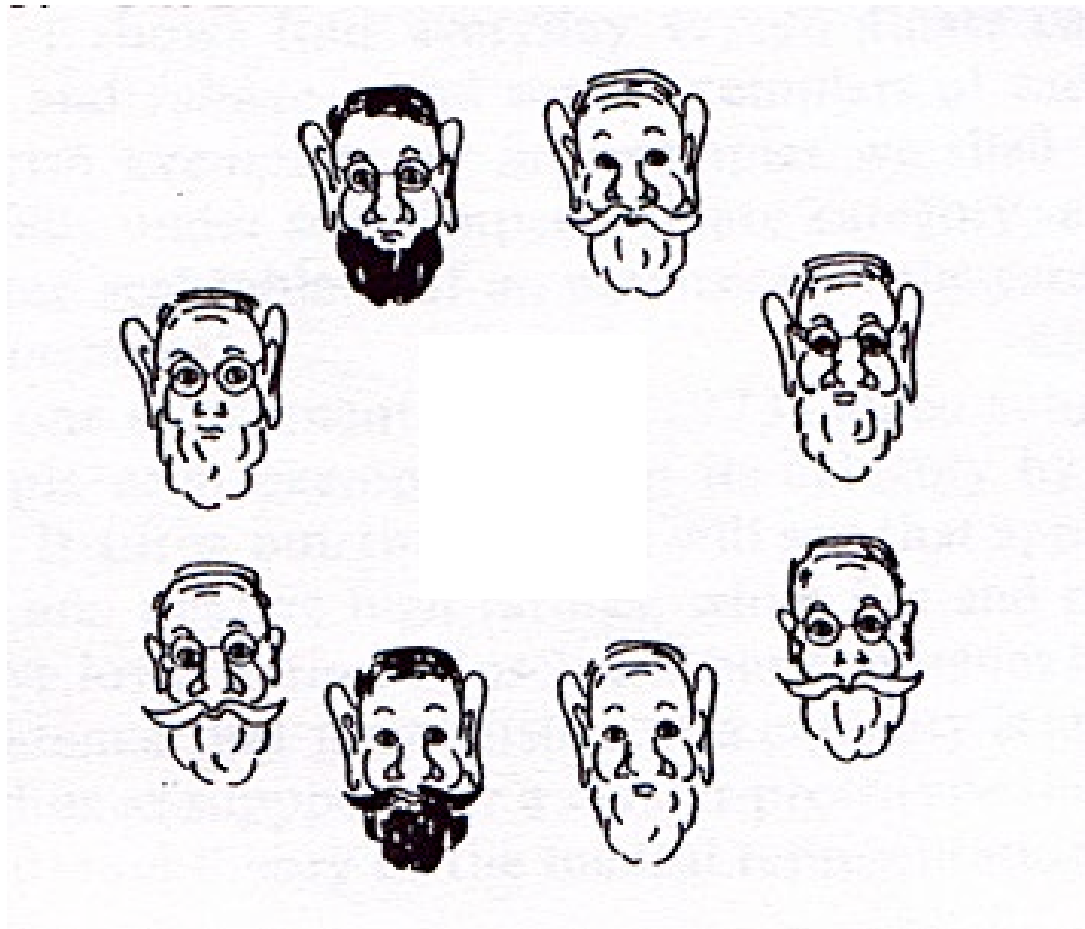
Approximating Bayesian inference

$$E_{p(h|d)}[f(h)] \approx \frac{\sum_i p(d | h^{(i)}) f(h^{(i)})}{\sum_i p(d | h^{(i)})}$$

Sample from the prior, weight by the likelihood

Does that equation remind you of any prior psychological theories?

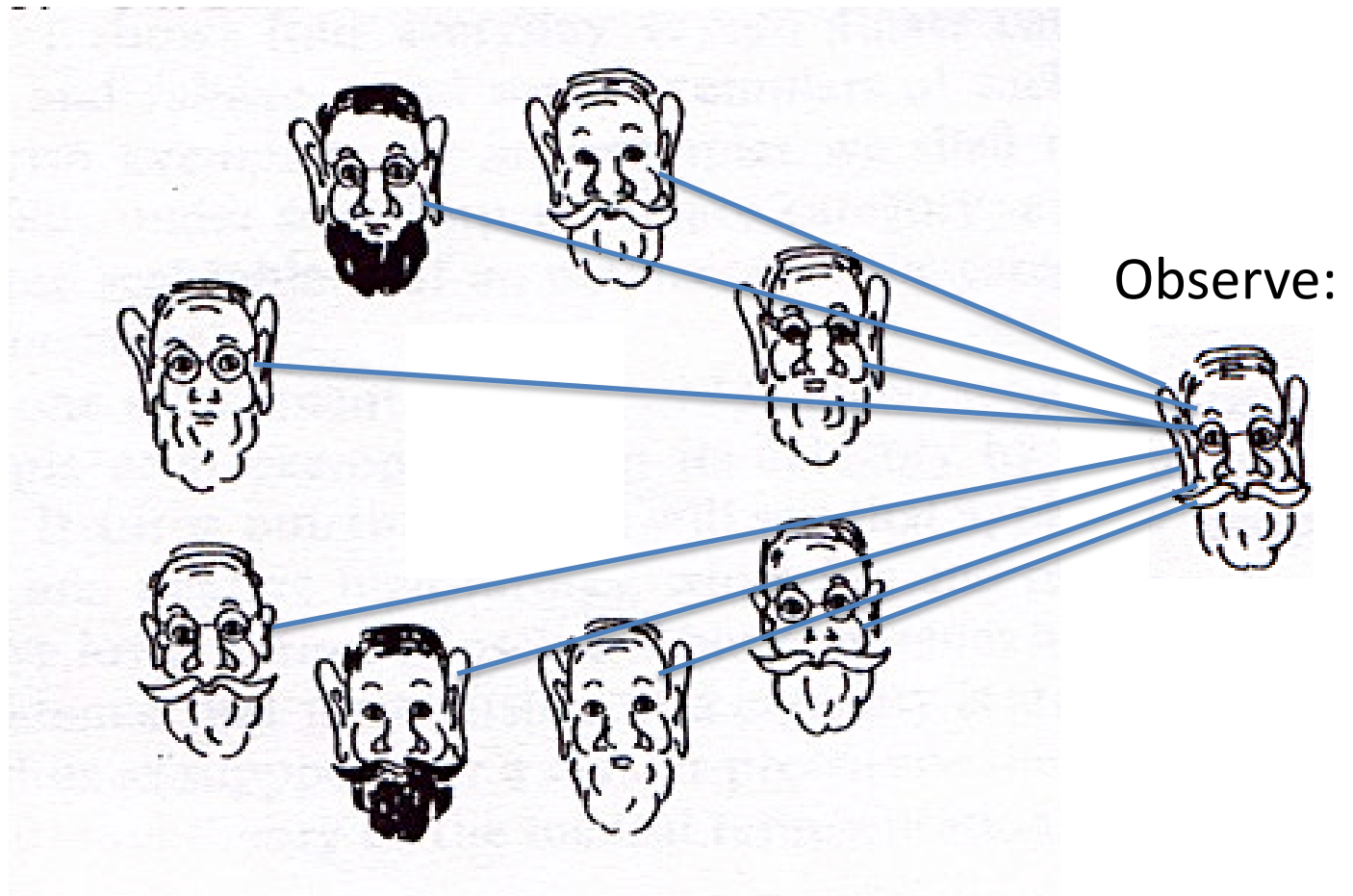
Exemplar theories



Store every member (“exemplar”) of the family

(Armstrong, Gleitman, & Gleitman, 1983)

Exemplar theories



Store every member (“exemplar”) of the family

(Armstrong, Gleitman, & Gleitman, 1983)

Exemplar models

Assume decisions are made by storing previous events in memory, then activating by similarity.

For example, categorization:

$$P_{choice}(c | x) = \frac{\sum_i s(x, x^{(i)}) I(x^{(i)} \in c)}{\sum_i s(x, x^{(i)})}$$

where $x^{(i)}$ are exemplars, $s(x, x^{(i)})$ is similarity, $I(x^{(i)} \in c)$ is 1 if $x^{(i)}$ is from category c .

(e.g., Nosofsky, 1986)

Exemplar models

Assume decisions are made by storing previous events in memory, then activating by similarity.

General version:

$$response(x) = \frac{\sum_i s(x, x^{(i)}) f(x^{(i)})}{\sum_i s(x, x^{(i)})}$$

where $x^{(i)}$ are exemplars, $s(x, x^{(i)})$ is similarity, $f(x^{(i)})$ is quantity of interest.

Equivalence

Exemplar: $response(x) = \frac{\sum_i s(x, x^{(i)}) f(x^{(i)})}{\sum_i s(x, x^{(i)})}$

IS: $E_{p(h|d)}[f(h)] \approx \frac{\sum_i p(d | h^{(i)}) f(h^{(i)})}{\sum_i p(d | h^{(i)})}$

When the samples are hypotheses, exemplar models can be viewed as importance samplers of Bayesian models.

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Updating distributions over time...

Computational costs are compounded when data are observed incrementally...

- recompute $P(h|d_1, \dots, d_n)$ after observing d_n

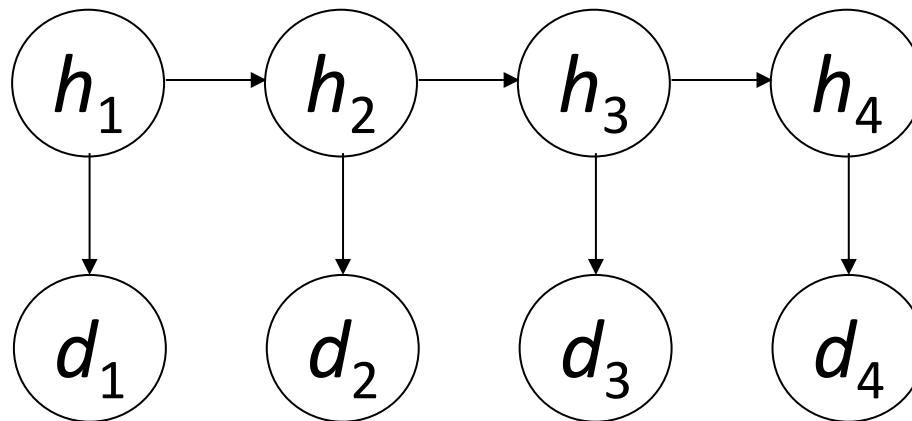
Exploit “yesterday’s posterior is today’s prior”

$$P(h \mid d_1, \dots, d_n) \propto P(d_n \mid h)P(h \mid d_1, \dots, d_{n-1})$$

Repeatedly using importance sampling results in an algorithm known as a “particle filter.”

Also called “sequential Monte Carlo,” “bootstrap filter,” “condensation,” and “survival of the fittest.”

Dynamic hypotheses



$$\begin{aligned} P(h_4 \mid d_1, \dots, d_4) &\propto P(d_4 \mid h_4) P(h_4 \mid d_1, \dots, d_3) \\ &= P(d_4 \mid h_4) \sum_{h_3} P(h_4 \mid h_3) \underbrace{P(h_3 \mid d_1, \dots, d_3)}_{w_3} \end{aligned}$$

Exploit this recursive relation to make a sampling algorithm!

w_n is the posterior after n steps.

Particle filter

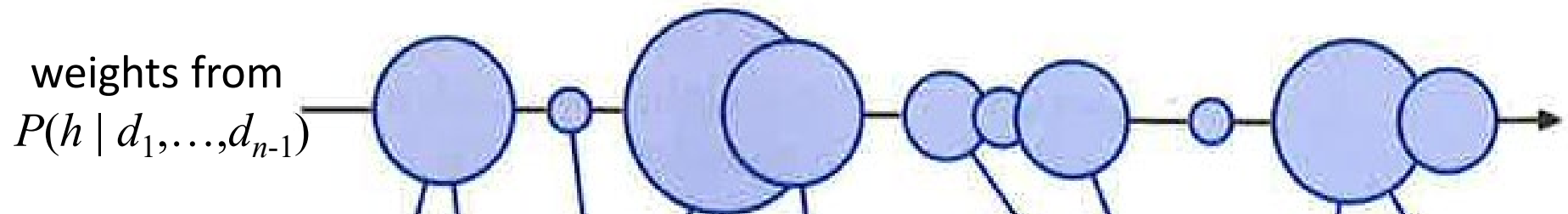
1. Initialize M samples (called “particles”) from prior and weights to be uniform over samples.

For data point $n = \{1, \dots, N\}$:

2. Update each sample’s weight by multiplying it by how probable the current datum is given the sample’s value.
3. Normalize the weights.
4. If the weights are becoming “degenerate,” resample (with replacement) particles according to current weight distribution. Reset weights to be uniform.

At end, samples approximate the posterior.

Particle filter



The promise of particle filters

A general scheme for defining rational process models of updating over time. (Sanborn et al., 2006)

Model limited memory, and produce order effects (cf. Kruschke, 2006)

Used to define rational process models of...

- categorization (Sanborn et al., 2006)
- associative learning (Daw & Courville, 2008)
- changepoint detection (Brown & Steyvers, 2009)
- sentence processing (Levy et al., 2009)

And more since 2009...

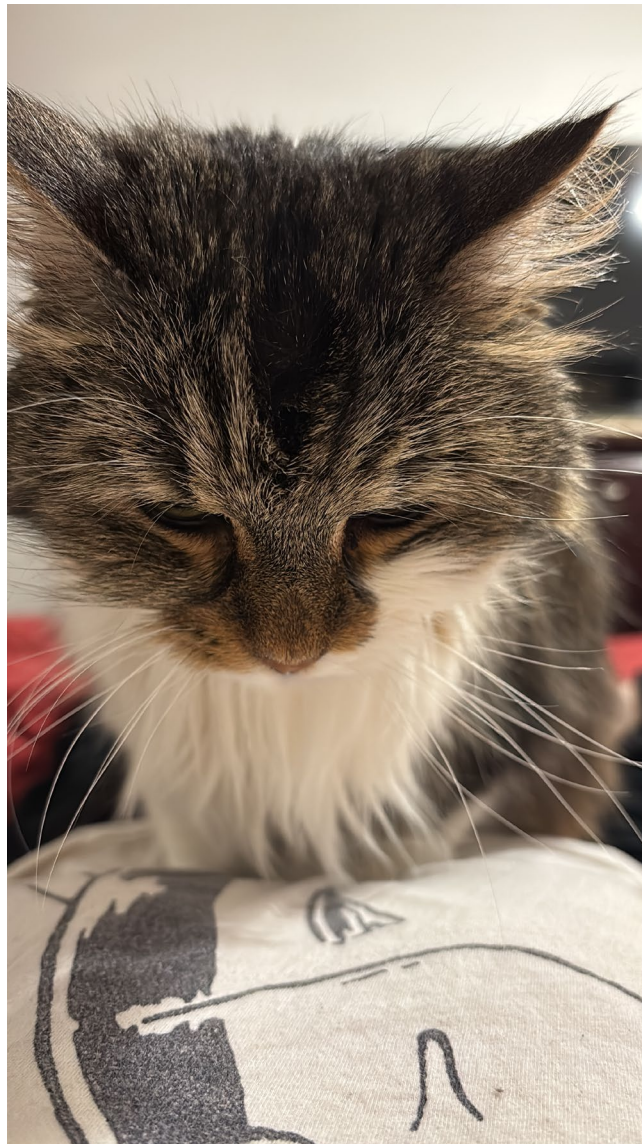
Want more on particle filters?

7 minute youtube video “Particle filtering without equations”:

<https://www.youtube.com/watch?v=aUkBa1zMKv4>

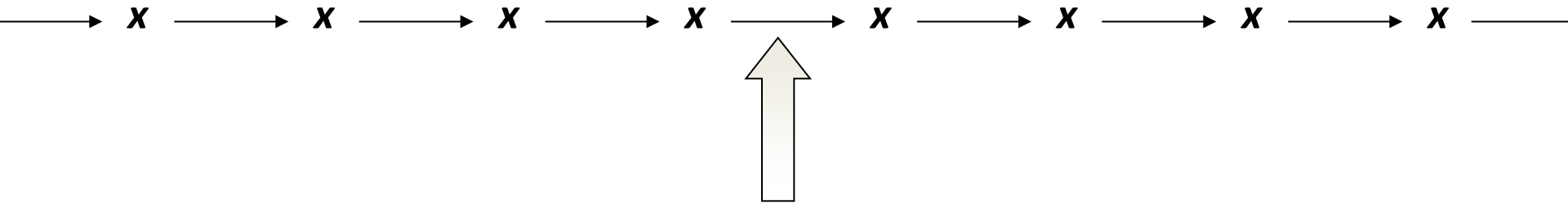
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Short rest 1



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Markov chains



Transition matrix

$$\mathbf{T} = P(\mathbf{x}^{(t+1)} | \mathbf{x}^{(t)})$$

Variables $\mathbf{x}^{(t+1)}$ independent of history given $\mathbf{x}^{(t)}$

Converges to a *stationary distribution* under easily checked conditions (i.e., if it is *ergodic*).

Markov chain Monte Carlo (MCMC)

Sample from a target distribution $P(\mathbf{x})$ by constructing Markov chain for which $P(\mathbf{x})$ is the stationary distribution

Two main schemes:

- **Metropolis-Hastings algorithm**
- Gibbs sampling

Metropolis-Hastings algorithm

(Metropolis et al., 1953; Hastings, 1970)

Step 0: Initialize values for the state of the chain

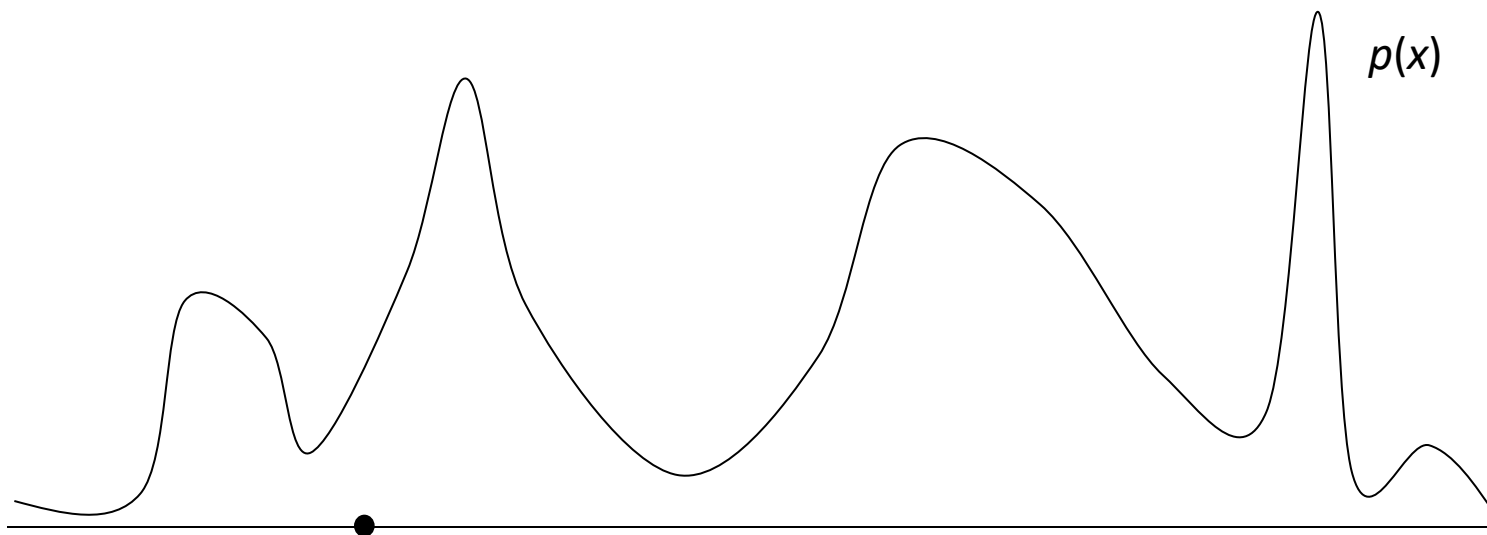
Step 1: propose a state (we assume symmetrically)

$$Q(x^{(t+1)} | x^{(t)}) = Q(x^{(t)} | x^{(t+1)})$$

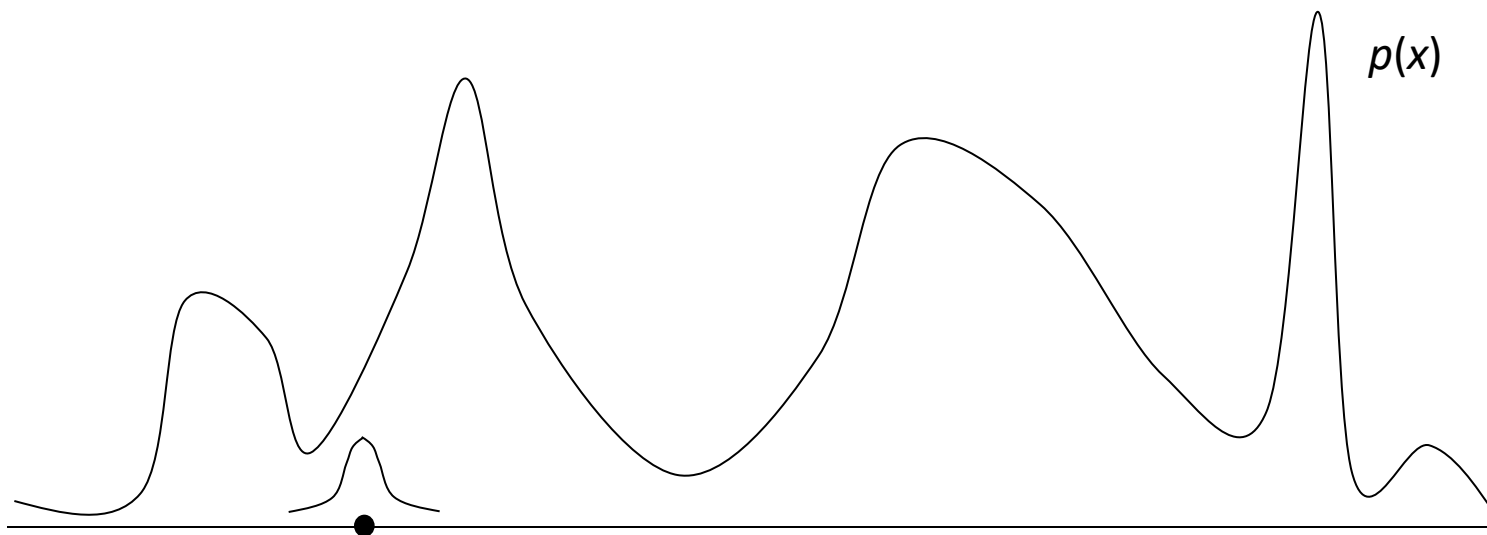
Step 2: decide whether to accept, with probability

$$A(x^{(t+1)}, x^{(t)}) = \min \left(1, \frac{p(x^{(t+1)})}{p(x^{(t)})} \right)$$

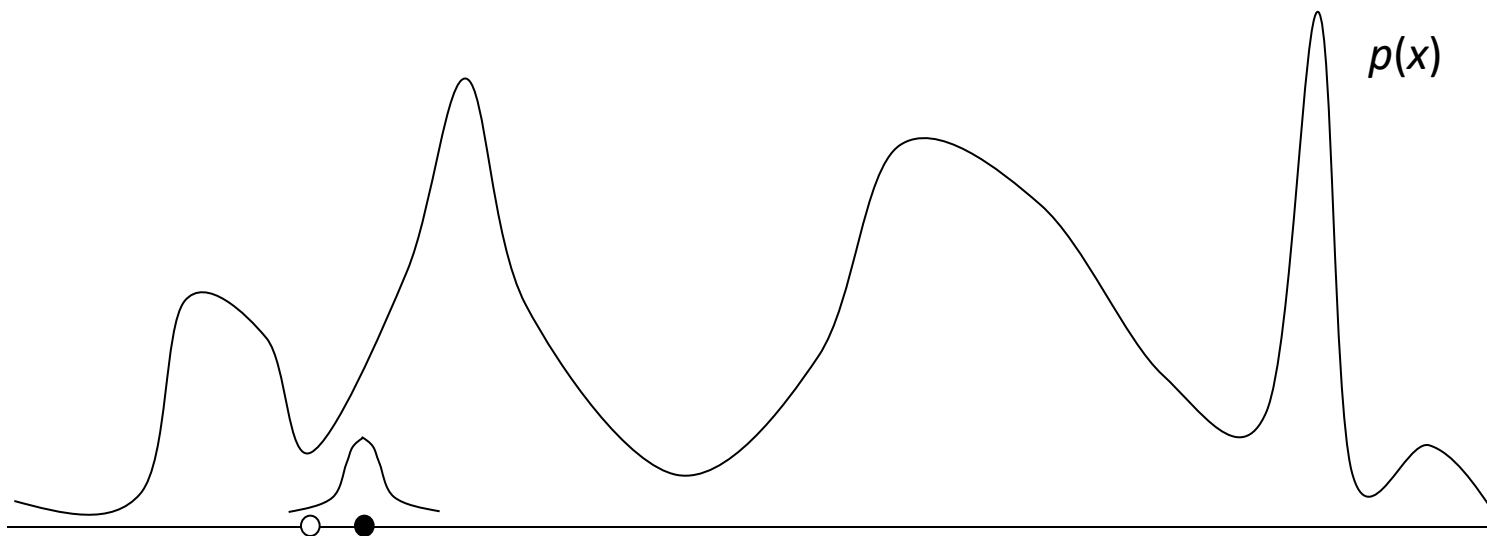
Metropolis-Hastings algorithm



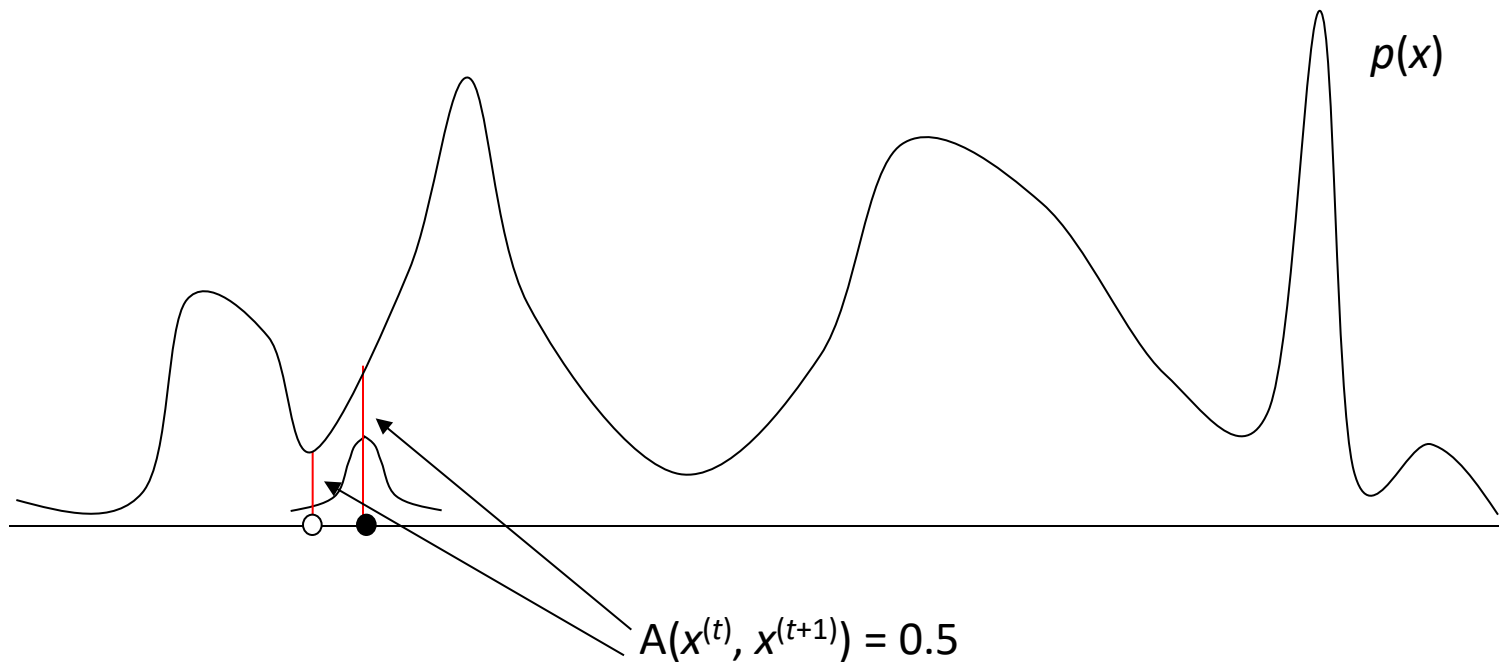
Metropolis-Hastings algorithm



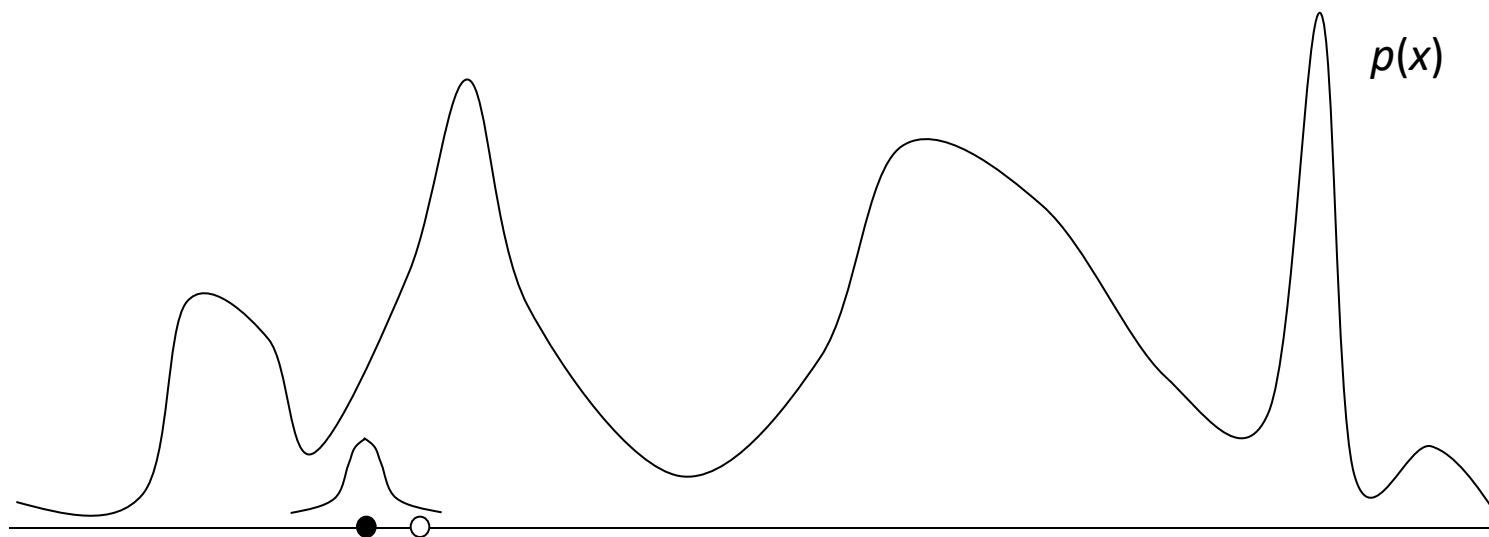
Metropolis-Hastings algorithm



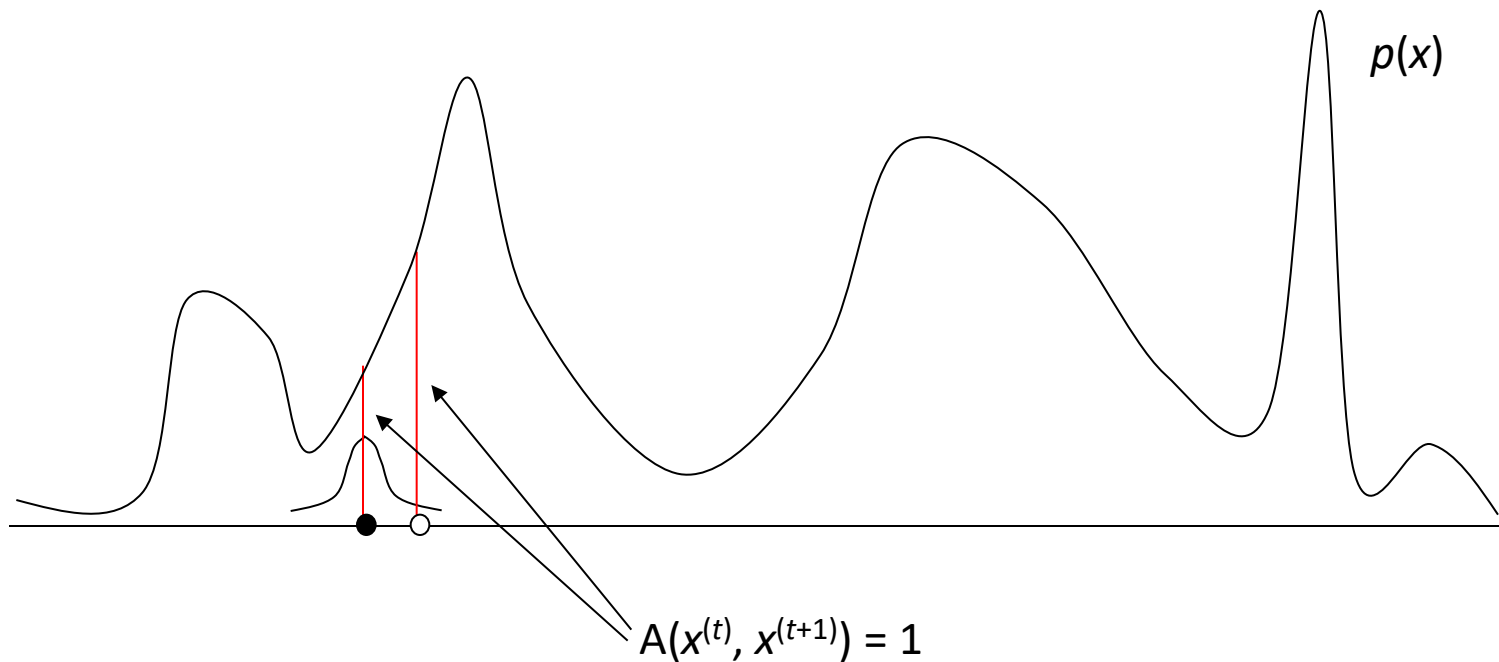
Metropolis-Hastings algorithm



Metropolis-Hastings algorithm



Metropolis-Hastings algorithm



Example

Visualizing Metropolis-Hastings/Gibbs Sampling

<https://chi-feng.github.io/mcmc-demo/>

<https://chi-feng.github.io/mcmc-demo/app.html?algorithm=RandomWalkMH&target=standard>

Markov chain Monte Carlo (MCMC)

Sample from a target distribution $P(\mathbf{x})$ by constructing Markov chain for which $P(\mathbf{x})$ is the stationary distribution

Two main schemes:

- Metropolis-Hastings algorithm
- **Gibbs sampling**

Gibbs sampling

1. Start a “chain” with a random state.
2. Pick a random variable in state.
3. Resample that variable’s value conditioned on the other
4. Go back to 2 and repeat indefinitely.

For variables $\mathbf{x} = x_1, x_2, \dots, x_n$ and target $P(\mathbf{x})$

Draw $x_i^{(t+1)}$ from $P(x_i | \mathbf{x}_{-i})$

$$\mathbf{x}_{-i} = x_1^{(t+1)}, x_2^{(t+1)}, \dots, x_{i-1}^{(t+1)}, x_{i+1}^{(t)}, \dots, x_n^{(t)}$$

Chain
Starts

$$x_1$$

$$x_2$$

$$x_3$$

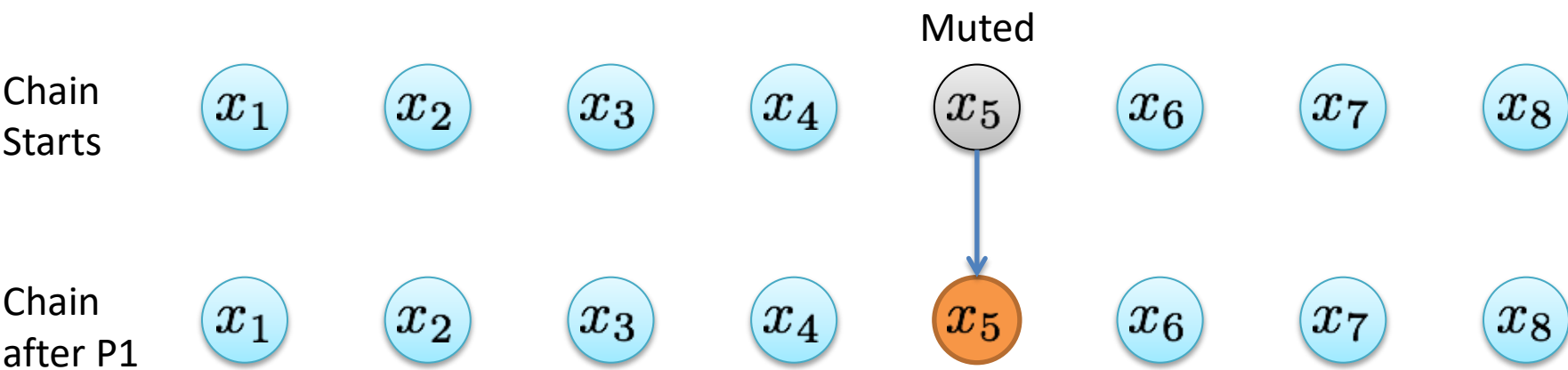
$$x_4$$

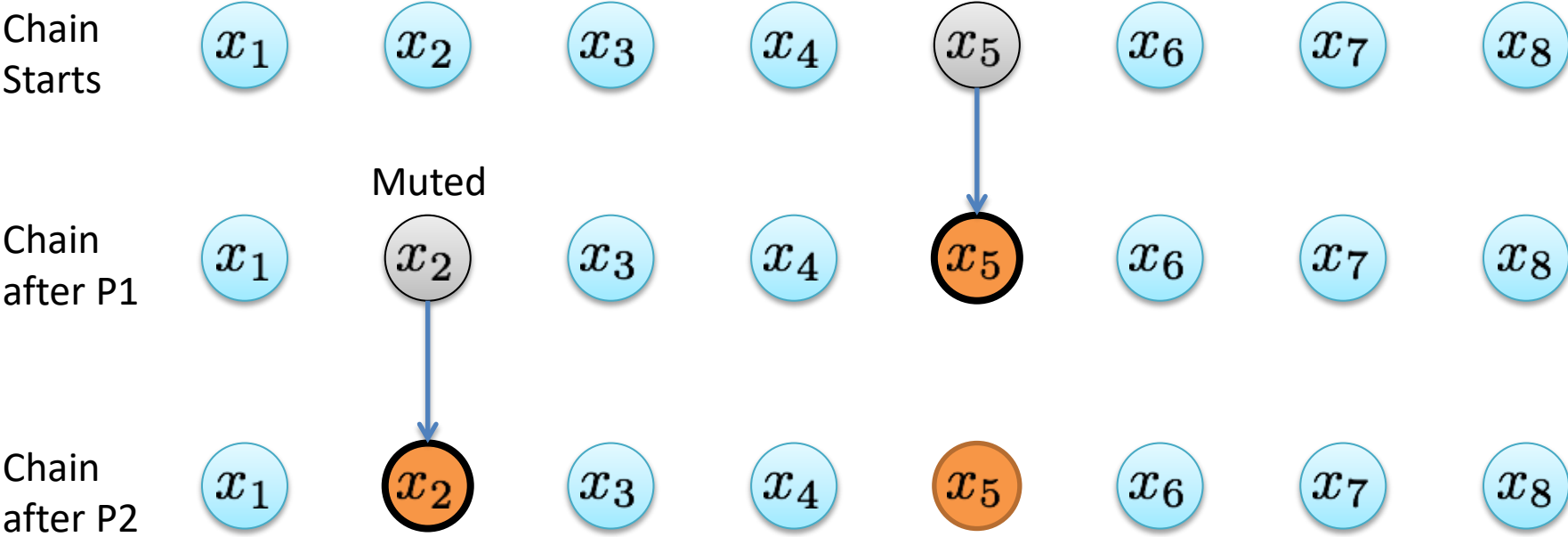
$$x_5$$

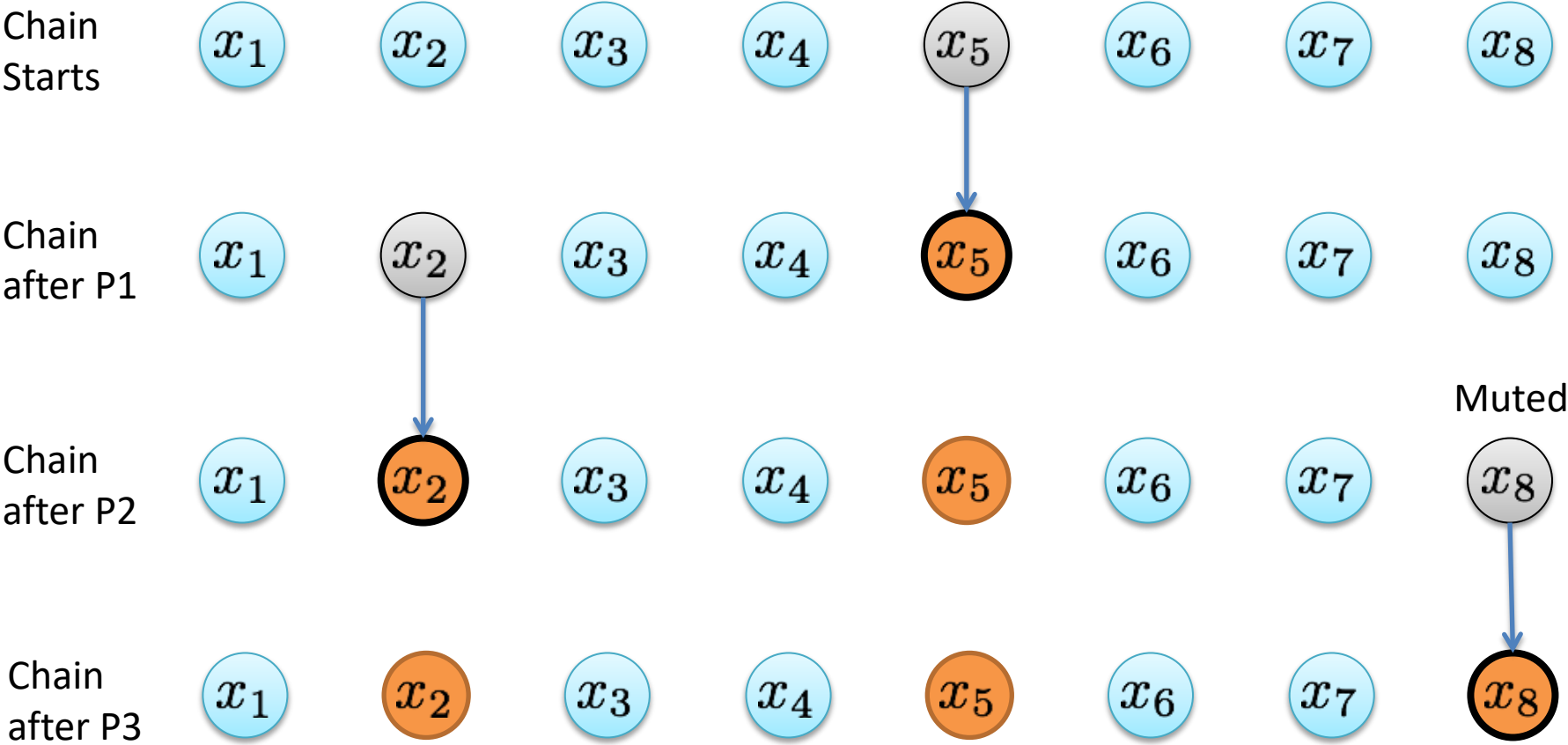
$$x_6$$

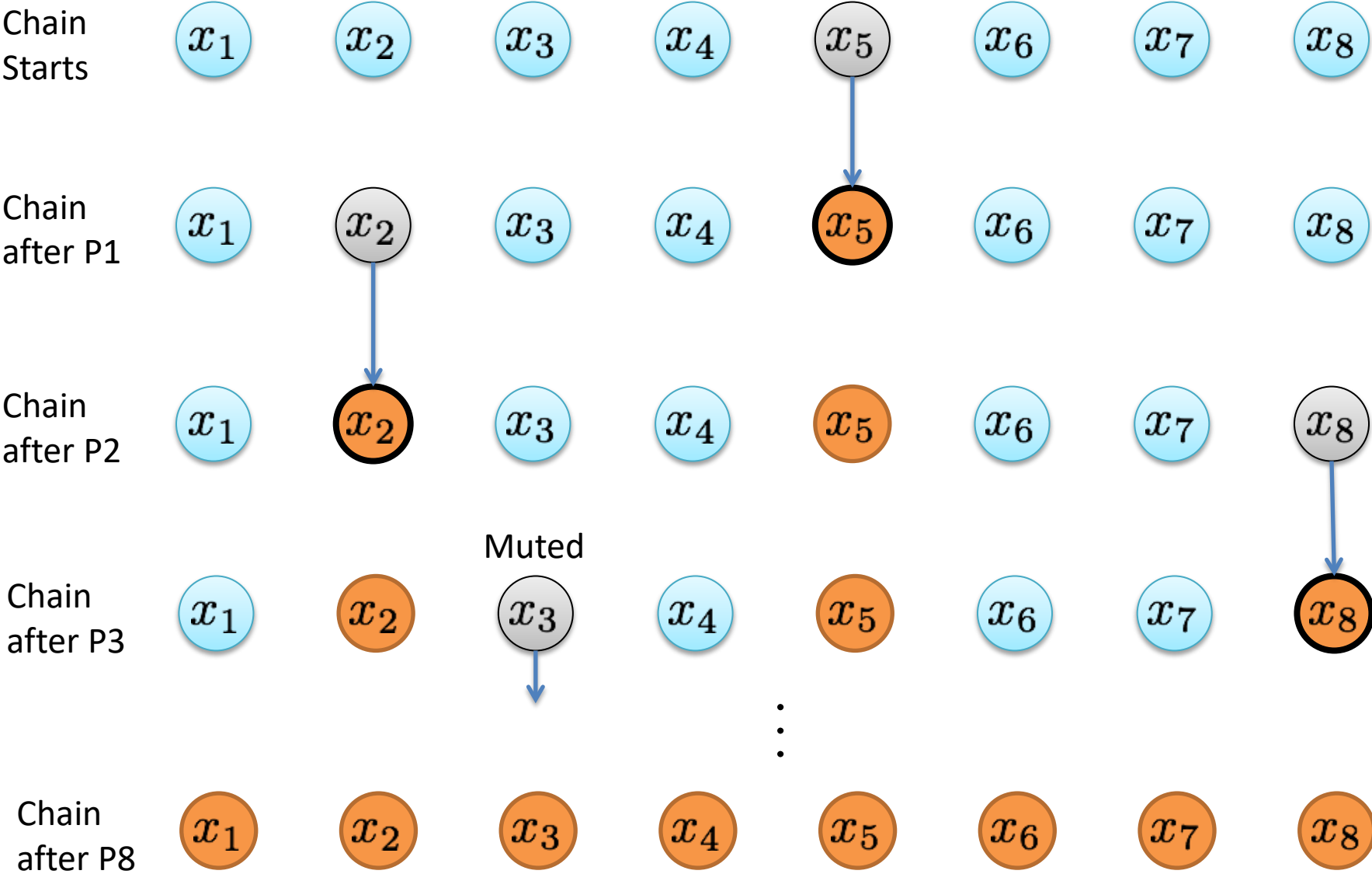
$$x_7$$

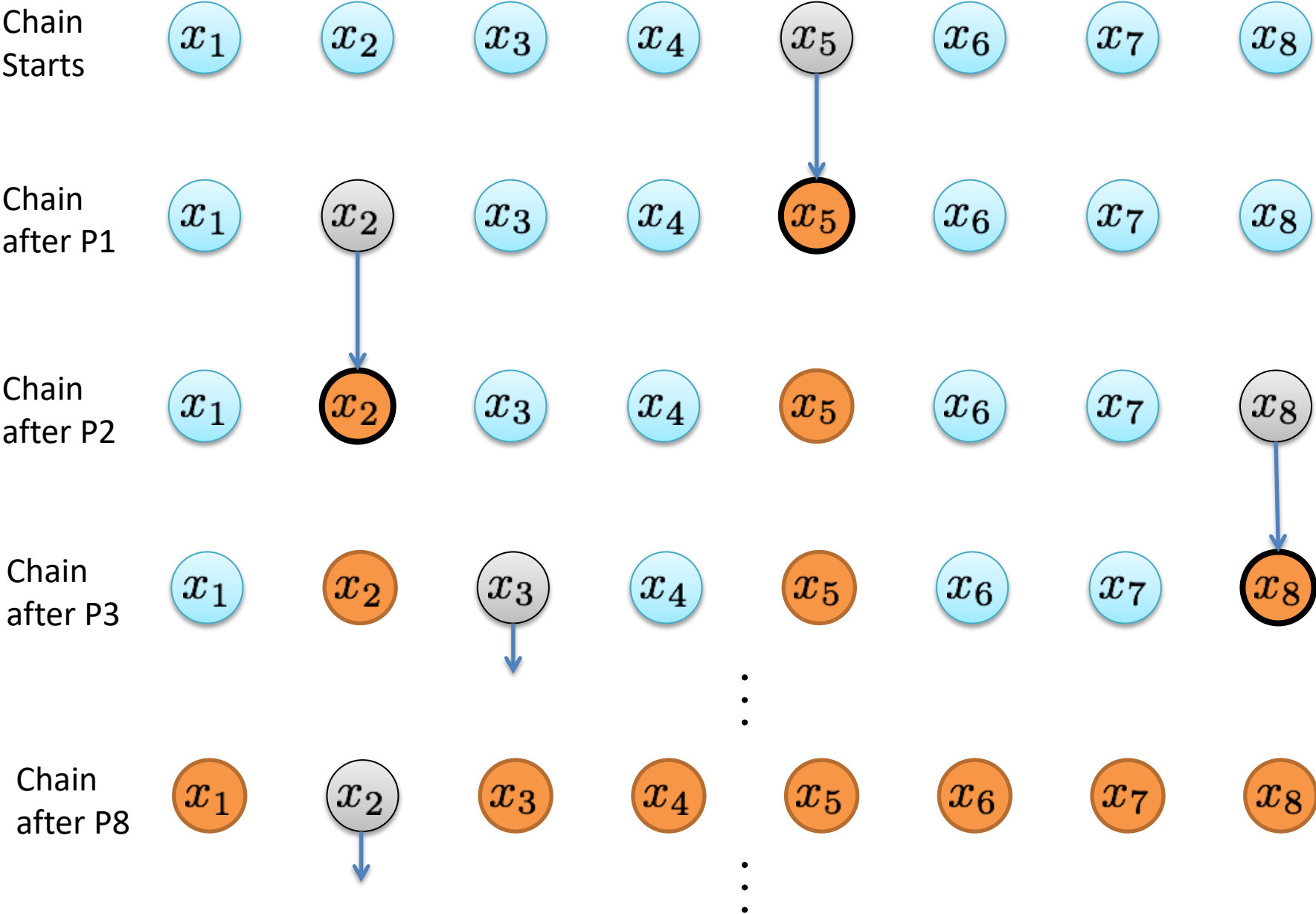
$$x_8$$











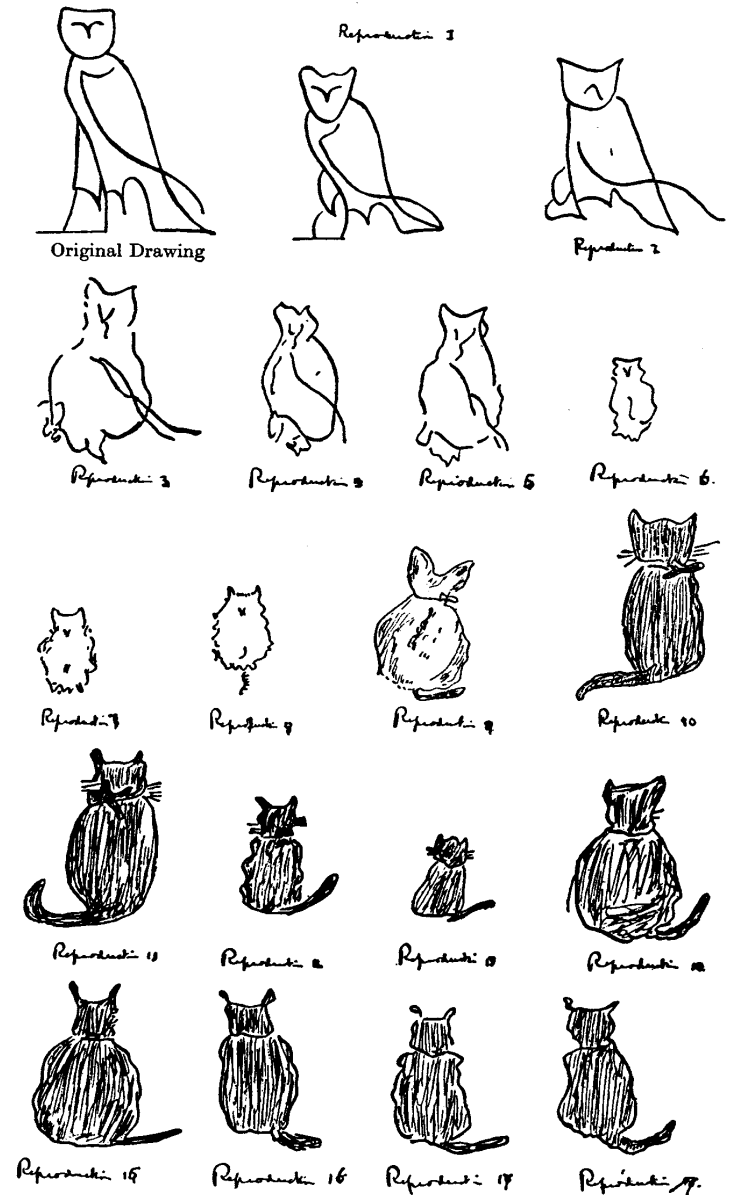
Serial reproduction

(Bartlett, 1932)

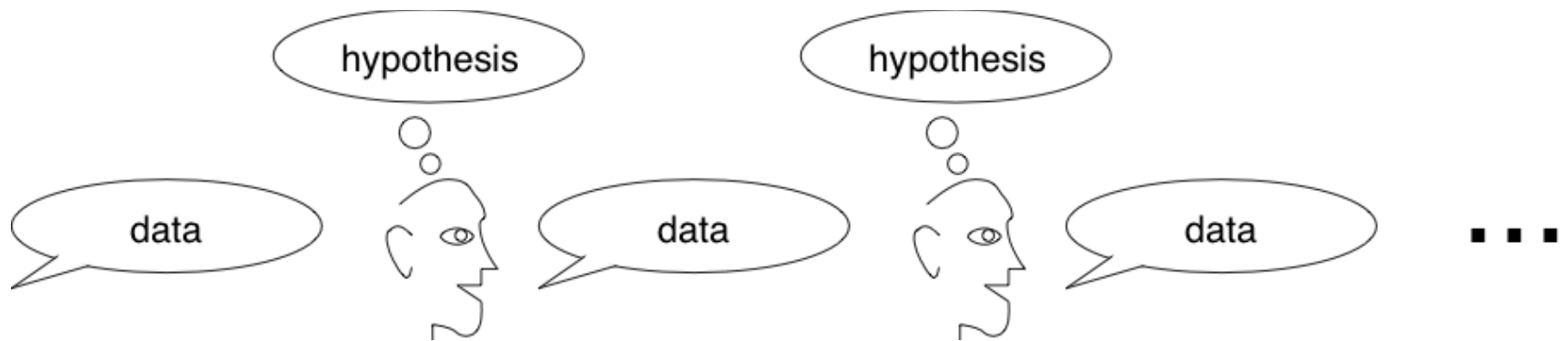
Participants see stimuli,
then reproduce them from
memory

Reproductions of one
participant are stimuli for
the next

Stimuli were interesting,
rather than controlled
– e.g., “War of the Ghosts”



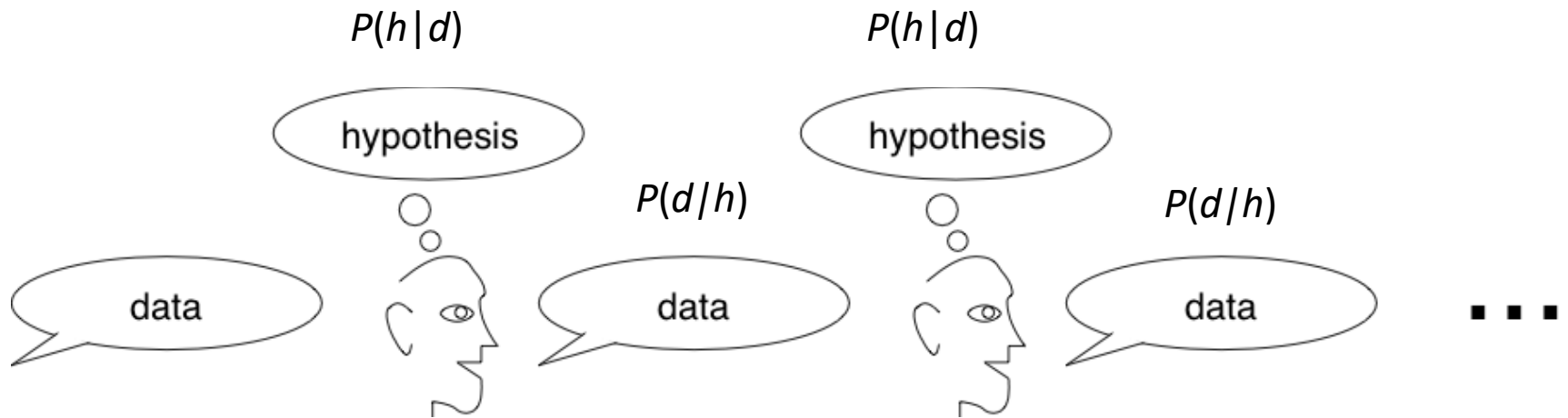
Human MCMC



What are the consequences of learners learning from other learners?

(Bartlett, 1932; Kirby, 2001; Griffiths & Kalish, 2007; ...)

Analyzing Human MCMC

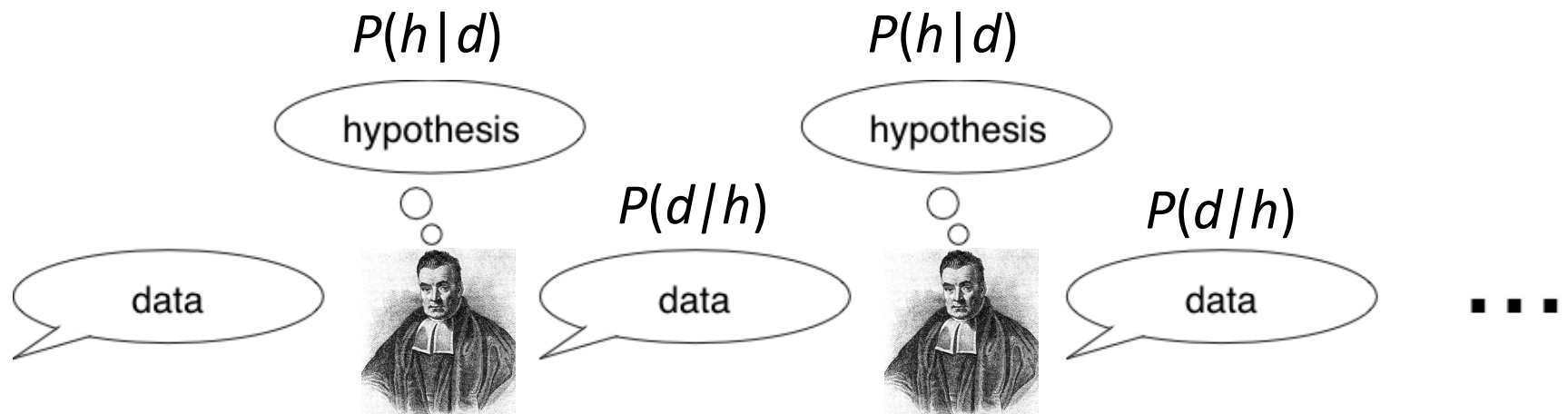


$P(h|d)$: probability of inferring hypothesis h from data d

$P(d|h)$: probability of generating data d from hypothesis h

Bayesian MCMC

(MCMC with People/"Iterated learning")



Assume learners *sample* from their posterior distribution:

$$P(h | d) = \frac{P(d | h)P(h)}{\sum_{h' \in H} P(d | h')P(h')}$$

(Bartlett, 1932; Kirby, 2001; Griffiths & Kalish, 2007; ...)

Analyzing Human MCMC

$$d_0 \xrightarrow{P(h|d)} h_1 \xrightarrow{P(d|h)} d_1 \xrightarrow{P(h|d)} h_2 \xrightarrow{P(d|h)} d_2 \xrightarrow{P(h|d)} h_3 \xrightarrow{\quad} \dots$$

A Markov chain on hypotheses -> Converges to prior on hyps. $P(h)$!

$$h_1 \xrightarrow{\sum_d P(d|h)P(h|d)} h_2 \xrightarrow{\sum_d P(d|h)P(h|d)} h_3 \xrightarrow{\quad} \dots$$

A Markov chain on data -> Converges to *prior predictive dist.* $P(d)$!

$$d_0 \xrightarrow{\sum_h P(h|d)P(d|h)} d_1 \xrightarrow{\sum_h P(h|d)P(d|h)} d_2 \xrightarrow{\sum_h P(h|d)P(d|h)} \dots$$

(Bartlett, 1932; Kirby, 2001; Griffiths & Kalish, 2007; ...)

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Short Rest 2

